
Complicity in Climate Harms: A Case Study of Australia's Gas Export Industry

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Australia is currently the world's second largest exporter of gas. The GHG emissions produced from exported gas add significantly to the risk of climate change. Yet according to current international conventions, fossil fuel-exporting States such as Australia are not liable for any of the harms to which their exported fossil fuels contribute. This article argues that the current "territorial" model for allocating responsibility for climate harms is inadequate and that fossil fuel-exporting States ought to be responsible for at least some of the harms to which their exported fossil fuel emissions contribute. Part II outlines the extent of Australia's gas export industry. Part III describes an account of complicity drawn from legal and moral philosophy and applies it to the case of Australia's gas export industry. Part IV discusses two ways in which gas use constitutes significant harm. The article closes by considering the policy implications of these moral arguments (Part V).

I. INTRODUCTION

Natural gas¹ is replacing coal as an energy source in many countries, and the global export market for gas is booming.² For gas-exporting countries such as Australia, the surge in gas consumption has had a significant impact on the economy. The Reserve Bank of Australia estimates that liquefied natural gas (LNG) exports began adding approximately 0.75 of a percentage point to Australia's gross domestic product from 2015.³ This is equivalent to around one-fifth of the 3.5% growth the Australian Treasury predicted would occur in that year. Australia will also soon be the number one exporter of gas in the world.⁴ In addition to its association with economic growth, gas has been identified by the recently released Finkel Report as an important transition fuel for Australia to reach its nationally determined emissions reduction target.⁵

This article argues that the boom in Australian exports of gas represents a problematic and inadequate response to climate change and ought to significantly increase Australia's liability for climate-related harms. There are, of course, many different reasons why increasing gas exports might be problematic. Some have pointed to the lack of proper taxation arrangements compared to other exporting countries.⁶ There are the well-known domestic supply constraints caused by selling gas into the international market.⁷ There are also concerns about environmental damage, for example through "fracking" for some

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¹ Throughout this article, "gas" refers to natural gas unless otherwise specified.

² International Energy Agency, *Short-term Energy Outlook* (2016) <<https://www.eia.gov/outlooks/steo/>>.

³ Natasha Cassidy and Mitch Kosev, "Australia and the Global LNG Market" (2015) *Reserve Bank of Australia Bulletin* 33.

⁴ Cassidy and Kosev, n 3.

⁵ Alan Finkel et al, *Blueprint for the Future: Independent Review into the Future Security of the National Electricity Market* (2017) 105–120 <<https://www.environment.gov.au/system/files/resources/1d6b0464-6162-4223-ac08-3395a6b1c7fa/files/electricity-market-review-final-report.pdf>>. The Finkel Report recommended a redoubling of efforts to prove the environmental sustainability of both conventional and non-conventional gas extraction methods.

⁶ International Transport Workers Federation, "Australian LNG Exports to Boom, Tax Revenue is a Bust" (ITF Briefing Paper, September 2016).

⁷ Australian Competition and Consumer Commission, *Inquiry into the East Coast Gas Market* (2016) <https://www.accc.gov.au/system/files/1074_Gas%20enquiry%20report_FA_21April.pdf>.

types of gas. These arguments are important and should form part of the overall assessment of the role of gas in combating climate change. However, the focus here is on moral arguments.

The central moral argument outlined here is that exporting large amounts of gas is a way of being *complicit* (ie indirectly involved) in serious harm to the climate. A consequence of this claim is that Australia, being a high fossil fuel-exporting country, ought to be held responsible for at least some of the greenhouse gas (GHG) emissions and associated harms that come with exporting large quantities of fossil fuels to other countries. Part II outlines the extent of Australia's gas export industry. Part III describes an account of complicity drawn from legal and moral philosophy and applies it to the case of Australia's gas export industry. Part IV discusses two ways in which gas use constitutes significant harm, relating to: (1) the fact that full life cycle GHG emissions from gas may be higher than expected due to methane leakage; and (2) opportunity costs associated with gas infrastructure. The article closes by considering the policy implications of these moral arguments (Part V).

II. AUSTRALIA'S GAS EXPORT INDUSTRY

Gas is a carbon-emitting fossil fuel primarily used for power generation, though it has a range of other industrial, commercial and residential uses. It reaches users either directly in its gaseous form through pipelines or via the production of LNG. Australia is currently the second-largest exporter of gas.⁸ By 2021, it will be the largest exporter of gas in the world, overtaking Qatar.⁹ The main export destinations are Japan, China, and South Korea,¹⁰ and Taiwan and India are important emerging markets. This development follows investment of approximately AU\$200 billion into the industry over the last decade.¹¹ In value terms, LNG was expected to become Australia's second-largest commodity export (after iron ore) by 2018.¹²

The boom in gas exports will also generate considerable revenue. It has been estimated, for example, that after 2020, the LNG industry will add \$30 billion a year to GDP, or about 2% of Australia's total GDP.¹³ After LNG plant construction is finished, between 55,000 and 65,000 people in total are expected to be employed in the wider LNG industry and its direct supply chain.¹⁴ Yet we should also note that tax revenue from these exports is likely to be limited. The International Transport Workers Federation¹⁵ estimate that in 2019–2021, both Qatar and Australia are forecast to export roughly 100 billion cubic metres of LNG, with the Government of Qatar to receive AU\$26.6 billion in revenue (in the form of royalties), while Australia's Commonwealth Government is expected to receive just AU\$800 million in revenue (in the form of petroleum resource rent tax (PRRT)).¹⁶

⁸ Holden Richard and Marieke D'Cruz, *Harnessing the Boom: How Australia Can Better Capture the Benefits of the Natural Gas Boom* (McKell Institute, April 2017) <<https://mckellinstitute.org.au/research/reports/harnessing-the-boom/>>.

⁹ Cassidy and Kosev, n 3.

¹⁰ Richard and D'Cruz, n 8.

¹¹ D Dediu et al, *Sustaining Impact from Australian LNG Operations* (2016) <<https://www.mckinsey.com/~media/McKinsey/Featured%20Insights/Asia%20Pacific/Sustaining%20impact%20from%20Australian%20LNG%20operations/Sustaining-impact-from-Australian-LNG-operations.ashx>>.

¹² Cassidy and Kosev, n 3.s

¹³ Joao Segorbe and Peter Lambert, *Meeting East Australia's Gas Supply Challenge* (McKinsey Australia, 2017).

¹⁴ Dediu et al, n 11.

¹⁵ International Transport Workers Federation, n 6.

¹⁶ The rationale for comparing Qatar royalties to Australian PRRT is that Australian PRRT was introduced to Australia to replace the Commonwealth royalty system in the 1980s. It should be noted that this comparison does not include royalties that Australian States receive for natural gas exports (between 10% and 12.5% of the wellhead value). While a State-based royalty figure for 2021 is not available, the total 2015–2016 financial year figure for onshore oil and gas projects was estimated at roughly AU\$700 million. Thus, it looks likely that, in 2021, Qatar revenue for natural gas exports will dwarf Australian figures, even if State-based royalties are taken into account: Australian Government, Department of the Treasury, *Petroleum Resource Rent Tax Review* (2017) <<https://cdn.tspace.gov.au/uploads/sites/72/2017/04/PRRT.pdf>>.

The volume of gas exported from Australia and the GHG emissions that it produces are huge. Table 1 below compares the total emissions that will be produced from Australian LNG exports against Australia's total domestic emissions for the equivalent years.

TABLE 1. Total Australian domestic emissions compared with carbon dioxide equivalent (CO₂-e) emissions from exported LNG.

	2014– 2015	2015– 2016	2016– 2017	2017– 2018	2018– 2019	2019– 2020	2020– 2021
Australian domestic emissions of CO ₂ -e	565.6	594.4	617.3	638.2	652.7	655.6	659.8
Australian CO ₂ -e emissions from exported LNG	67	95.95	135.88	182.51	198.32	199.13	201.54
Australian emissions including exported LNG	632.6	690.35	753.18	820.71	851.02	854.73	861.34

Notes. All figures are in Mt CO₂-e. Australian domestic emissions figures are taken from the Department of the Environment's *Australia's Emission Projections 2029–2030* (March 2015) 32. Export figures are taken from the Office of the Chief Economist, *Resources and Energy Quarterly* (March 2016) 79. Conversion factors used are given in the Australian Government's *Greenhouse Gas Accounts Factors* (August 2016) 12.

The first row of Table 1 details actual total Australian domestic emissions from consumption of fossil fuel resources. The second row lists the CO₂-e emissions that will be produced from combustion of the LNG exported by Australia across the same period of years. The third column provides the sum of these two figures. In the financial year 2014–2015, LNG export emissions were equivalent to approximately one-sixth of the Australian domestic emissions figure. However, by the financial year 2020–2021, LNG export emissions will be approaching one-third of the Australian domestic emissions figure. In this year, the sum of LNG export emissions and domestic consumption emissions will be 861 Mt of CO₂-e. This figure is over one-third higher than Australia's emissions target for the same year, which is 532 Mt CO₂-e.¹⁷

These figures show that gas produced and exported by Australia leads to a significant quantity of emissions, especially when considered in relation to Australia's domestic emissions. Australia also has considerable remaining gas reserves, with a conservative estimate of CO₂-e emissions associated with such reserves sitting at 14,419.07 Mt CO₂-e.¹⁸ The scale of actual and projected emissions, combined with the possibility that Australia will continue utilising its remaining reserves well into the future, raises questions concerning the level of responsibility Australia ought to have for export-related emissions and the climate harms to which they contribute.

III. COMPLICITY IN HARM

As shown in Table 1, Australia's exported gas has contributed amounts equal to approximately one-third of the total volume of Australian domestic emissions from 2017. Yet, under current accounting rules States and the companies that operate within them, are not held accountable for the emissions produced from these exports. Yet, what we can call the "territorial model" of emissions accounting does not capture the moral responsibility of States.

With the exception of air and sea travel, most of the GHGs are produced by people within the territory of States, so when it comes to assigning responsibility for emissions and the harm to which they might have

¹⁷ Australian Government, Department of the Environment and Energy, *Australia's Emissions Projections 2029–2030* (2015) <<https://www.environment.gov.au/system/files/resources/f4bdfc0e-9a05-4c0b-bb04-e628ba4b12fd/files/australias-emissions-projections-2014-15.pdf>>.

¹⁸ The gas reserves figure is based on Geoscience Australia's estimates of total identified gas resources in Australia. Conversion factors used are given in the Australian Government's *Greenhouse Gas Accounts Factors*: Geoscience Australia, Australian Government, *Gas: Summary* (August 2016) 12 <<http://www.ga.gov.au/aera/gas>>.

contributed, the task ought to be fairly simple: countries ought to take responsibility for those emissions and any associated harms, to which they have contributed within their territorial borders. This kind of approach is utilised by the *United Nations Framework Convention on Climate Change* (UNFCCC) when it determines national (domestic) inventories of GHG emissions as part of the discussion of the global “carbon budget”. The global carbon budget represents the finite amount of GHGs that humankind can emit, consistent with an acceptable likely rise in global temperatures. According to the UNFCCC guidelines “greenhouse gas emissions and removals taking place within national territory and offshore areas over which the country has jurisdiction” (1.4)¹⁹ are included within that State’s carbon budget.

What the UNFCCC calls “scope 1+2” emissions are those emissions that are produced within a State’s borders by various types of activity, such as industrial activity, transport, and so on. Scope 3 emissions are the emissions that are produced by the commodities when they are generated outside the State’s territory and these are not part of the State’s emissions budget. To give an example, if a State exports coal, the emissions that are generated by extracting the coal and transporting it to a port are part of that State’s emissions budget because they occur within its territorial boundary. The emissions that are produced when the coal is burnt are part of the actual budget of the State that burns them. Including emissions in a State’s budget is one way of taking responsibility for emissions but not the only way. A State could assume financial obligations to pay the costs of adaptation or mitigation or compensate others for harms done. Nonetheless, the budget model above serves to illustrate how responsibility for emissions is determined.

If the formula set down by the UNFCCC in counting emissions is applied to liability for harm from emissions, States exporting fossil fuels are not deemed responsible for the harms that may result from the emissions produced from burning them. However, there is a plausible moral case for claiming that States have responsibility for at least part of the harms caused by their scope 3 emissions as well as a duty to stop causing those harms.

To see how this might be the case, let us bracket climate issues and consider the following analogies. Suppose that a country produced and exported large amounts of tobacco to a developing State that did not have health warnings for smoking. Given what we know about the links between smoking and death and disease, the exporting State is implicated in the harm caused, morally responsible for that harm and have a duty to take on at least some of the costs of that harm. Another example concerns hazardous waste. Where one State knowingly ships its dangerous medical or industrial waste to a country that has low or no standards for its safe disposal, we can say that the exporter bears some responsibility for harms that may result when the waste is not disposed of properly. This is likely to be the case even where there was consent from the importer to take the waste.

More obvious still is the case of uranium exports. There are good reasons why many States place restrictions on the end destinations of their product. States place restrictions on the sale of uranium because the risks of weapons proliferation, accidents at reactors, storage issues, and so on, are just too great with some countries to countenance export programs. Countries such as Australia are bound by the Nuclear Non-Proliferation Treaty to only sell uranium for peaceful uses.²⁰ Should one State knowingly export uranium to another State where safety is lax, it could rightly be accused of being irresponsible and having a share in the blame if an accident were to happen.

What all three of the cases above have in common is that the exporters cause harm in morally significant and blameworthy ways. The harm caused by GHG emissions are like the harms caused by tobacco in morally relevant respects: exporting countries contribute to knowingly harming the significant interests of others. For a State to fully accrue liability from the harms to which it contributes by selling fossil fuels such as gas, it must make a contribution to a harm, affect significant interests, do so knowingly and having alternatives. If this is true, the question is whether Australia meets these criteria in the case of gas exports. The second condition is obviously met given the significant interests at stake. The fourth condition is also met, as a wealthy country Australia has alternatives to generating its income in this way.

¹⁹ HS Eggleston et al (eds), *IPCC Guidelines for National Greenhouse Gas Inventories* (IGES, 2006).

²⁰ United Nations Office for Disarmament Affairs. *The United Nations and Disarmament – UNODA* (United Nations Office for Disarmament Affairs, 1970) <<https://www.un.org/disarmament/publications/yearbook/volume-1945-1970/>>.

In terms of the second condition, whether a country such as Australia had knowledge of the likely harms that might result from its exports (the knowledge condition), a plausible standard is that it must know or should know that its actions are contributing or are likely to contribute to a harm. All that is plausibly required for liability is that an agent knowingly contributes to the potentially harmful action or is ignorant (but should not have been), and that they know that their action is a contribution to the harm committed by someone else. Even if the countries that emit the GHGs by burning the fossil fuels intend to do so and the exporting country has no such intention, this ought not to absolve the exporter of blame because the exporting country should have known what the end use of the fossil would be. Knowing or being in a position where one should have known is a crucial part of contributing to harm.

Turning to the second condition, how does an exporter contribute to harm? After all, in the climate case countries or individuals can be involved in contributing to harmful outcomes either indirectly or directly. States contribute directly via the emissions produced within their own borders. But States can contribute to causing emissions indirectly, for instance by: supplying other emitters with fossil fuels, supporting policies that increase emissions or that prevent low emission alternatives, provide subsidies to emitters, influence the political process in favour of high-emitting companies, supply essential resources to emitters, or stall global climate treaties, and so on.

But if the main type of contributions to climate change from gas exporters are indirect contributions, how are we to describe and assess the moral liability of contributing to harm in this indirect way, especially in comparison to more direct contributions? A plausible framework for explaining how exporting fossil fuels is harmful is via the language of complicity. Typically, in legal and moral theory, an agent is complicit when they knowingly assist or encourage the harms of others.²¹ Through involvement in the harms done directly by primary or principal agents, a secondary agent becomes an accomplice or accessory and, according to this formula, shares some of the liability for the harms done. The badness of the harm committed by the secondary agent derives from the harm of the primary agent. A secondary agent can be complicit in different ways including: actively conspiring and planning a harm with a principal agent, simply co-operating with them, encouraging, permitting, soliciting or by aiding and abetting.

It is important to note the complexity that can be involved in being complicit in increasing the risk of climate harms through exporting fossil fuels. For example, a State could be complicit through directly *supplying* fossil fuels (being an “upstream producer”). The second way is through the various types of *support*. Support might take the form of providing the physical infrastructure that allows gas and other fossil fuels to be sold: the ports, roads, rail lines, and so on. A second dimension of support is the financing and subsidies that supports production. Subsidies to fossil fuel companies and products can often be difficult to categorise. Subsidies are typically thought of as being of three types: consumer, production or post-tax (externalities). Consumer subsidies typically take the form of regulating fuel prices or cash handouts to different groups of consumers. Production subsidies are directed at the fossil fuel production process itself and might include, tax breaks, cheap public finance, deductions for research and development or explorations investment or indemnities for environmental hazards. Some methodologies include post-tax costs (externalities), such as the cost of pollution, which dramatically increases the estimates of the extent of fossil fuel subsidies. For instance, an IMF report incorporating externalities into its methodology estimated that in 2015 alone energy subsidies globally (mostly fossil fuels) were \$5.3 *trillion* dollars.²² Other estimates range from \$160 billion to \$440 billion per year.²³

Leaving aside the merits of the different ways of calculating subsidies, what is important about each of these types of subsidies is that they provide crucial *benefits* for the fossil fuel industry, which is why subsidies are such a crucial component of support. The key feature of subsidies that ought to be of moral concern is the fact that they confer a benefit on a recipient relative to other recipients, whether the latter are individuals or specific industries. Indeed, this is how the World Trade Organization (WTO)

²¹ Chiara Lepora and Robert E Goodin, *On Complicity and Compromise* (OUP, 2013); Christopher Kutz, *Complicity: Ethics and Law for a Collective Age* (CUP, 2007).

²² D Coady, et al. *How Larger are Global Energy Subsidies?* (IMF working paper, 2015).

²³ E Bast, et al. *Empty Promises: G20 Subsidies to Oil, Gas and Coal Production* (Overseas Development Institute /Oil Change International, 2015) <<https://www.odi.org/publications/10058-empty-promises-g20-subsidies-oil-gas-and-coal-production>>.

has understood the significance of subsidies. WTO defines a subsidy as “any financial contribution by a government, or agent of a government, that is recipient-specific and confers a benefit on its recipients in comparison to other market recipients”.²⁴ There are, in fact, a huge array of measures that might count as subsidies. A recent report by the Overseas Development Institute (Bast et al, 2015) lists 30 different types. In the case of Australia, recent reports by the IMF and Oil Change estimate that Australia spends around AU\$5 billion pa on support of one kind or another.²⁵

A plausible third category of contribution to emissions could be called *influence*. Governments, powerful lobby groups or companies themselves can influence decisions about energy policy that favour fossil fuels. This can occur via trade arrangements, fossil fuel favourable domestic legislation, securing subsidies to make coal competitive, weakening global climate agreements, providing finance for coal-related infrastructure or in general via influencing governments to favour fossil fuel production or use. Fossil fuel companies themselves often engage in this kind of influence when they want a government to adopt a particular policy.

These three indirect ways of contributing to the risk of harm – supply, support and influence – are what makes fossil fuel-exporting States complicit. Through supplying, supporting and influencing emission outcomes Australia is complicit in risking harming others. Of all the components that go into providing gas to a buyer the act of selling gas is the most proximate to the harm itself. Nonetheless, without the types of indirect support listed above it is also less likely that the fossil fuel would not have been produced or sold because the various forms of support benefit the fossil fuel industry making their products easier to produce or sell. If States are complicit in harming others in this way via their support of exports then it provides States with a *negative duty* – a duty not to interfere or harm others – to stop their involvement in this harm in proportion to their involvement in contributing to the harm.

In regard to influence, we should note that by actively planning and conspiring with primary agents such as power companies, a country may be more than just a secondary agent – it is a principle or co-principals. Co-principles do more than just contribute to a harm, what they do constitutes the harm. Where countries act together to influence decisions or alter outcomes, they are co-operating (planning and executing a wrongdoing together) or colluding (acting in secret to deceive others and obtain a benefit). In these cases, in addition to supplying fossil fuels in a way that is complicit with the actions of a primary agent, they are also co-principals in virtue of their actions in co-operating or colluding to bring about a harmful act.

IV. THE HARMFULNESS OF GAS CONSUMPTION

So far, we have argued that countries are complicit in harm when they export large amounts of gas to other countries. However, one potential objection to this argument might be found in the claim that gas is a “transition” fuel towards a cleaner energy future because it produces substantially less emissions than other fossil fuels when combusted.²⁶ If gas does indeed produce substantially less emissions than other fossil fuels, perhaps it is reasonable to claim that the extent of the harm associated with a large gas exporting industry is less significant than might otherwise be assumed. But this claim is not straight forward. Here we discuss several reasons why gas might not be as climate friendly as is often claimed.

A. The Bare Harm of GHG Emissions from Gas on Global Warming

We should note that emitting *any* amount of GHG emissions potentially contributes to the risk of dangerous climate change in a “bare” sense, that is, in the sense that GHG emissions from gas contribute to the risk of climate change. As such, gas consumption may contribute to climate change, because gas burned for energy releases GHG emissions. However, there has been some debate about the extent of

²⁴ WTO quoted in Overseas Development Institute and Oil Change International, *Empty Promises: G20 Subsidies to Oil, Gas and Coal Production* (2015) 32.

²⁵ See n 22.

²⁶ David Biello, *Fact or Fiction?: Natural Gas Will Reduce Global Warming Pollution* (2015) Scientific America <<https://www.scientificamerican.com/article/fact-or-fiction-natural-gas-will-reduce-global-warming-pollution/>>; International Energy Agency, *World Energy Outlook 2017* (2017) <<https://www.iea.org/weo2017/>>; ProPublica, Duke Energy, *Just Hot Air?: Are Natural*

the GHG emissions associated with gas consumption, especially in terms of how gas fares compared to other fossil fuels.

As noted above, we know that the combustion of gas produces less emissions than other fossil fuels such as coal and oil.²⁷ However, Howarth has provided modelling estimates suggesting that gas is more harmful than is otherwise assumed when taking full life cycle GHG emissions into account.²⁸ Howarth et al.²⁹ describe a global warming potential (GWP) approach comparing emissions from gas and other fossil fuels over a 20-year and a 100-year time period. Across a 20-year time period, shale gas produced more GHG than coal or oil even using “low-end” estimates for methane emissions.³⁰ In addition, conventional gas also produced more GHG than coal or oil using both “medium” and “high-end” methane emissions estimates. However, conventional gas did not produce more GHG than coal at the very “low-end” range for methane emissions (ie the low emissions scenario). Across the 100-year time period, the influence of methane was much lower, yet using “high-end” methane emissions estimates, the GHG produced by both shale gas and conventional gas still exceeded that of coal and oil. This work suggests that full life cycle emissions from gas are greater than is often assumed.

The evidence from Howarth et al is less damning for gas on a 100-year rather than a 20-year timeframe. Indeed, whether a 100-year or a 20-year timeframe is more appropriate is a matter of contention in current debates.³¹ However, one reason to think that a 20-year time scale is relevant to the risk of dangerous climate change is that a temperature spike (even if it is a short-term one) may risk a fundamental change in the Earth's climate system, a tipping point that distinguishes climate change from dangerous climate change.³² For example, a short-term temperature spike may risk realisation of multiple mechanisms for such catastrophic change, such as the melting of methane clathrates in the oceans and melting of permafrost in the Arctic.³³

At any rate, further evidence that gas is not as climate-friendly as is sometimes assumed can be found independently from timescale considerations. To avoid the need to make a choice about what global warming timeframe to use, Alvarez et al.³⁴ adopted the technology warming potential method, which calculates atmospheric pressure as a function of time. Using this method, Alvarez et al compared gas

Gas's Environmental Benefits Overstated? (Scientific American, 2014); AR Brandt, et al. “Methane Leaks from North American Natural Gas Systems” (2014) 343 *Science* 733–735; Alan Finkel, *Independent Review into the Future Security of the National Electricity Market* (Australian Government Department of the Environment and Energy, 2016) <<https://www.energy.gov.au/government-priorities/energy-markets/independent-review-future-security-national-electricity-market>>.

²⁷ International Energy Agency, n 26.

²⁸ Robert W Howarth, “A Bridge to Nowhere: Methane Emissions and the Greenhouse Gas Footprint of Natural Gas” (2014) 2 *Energy Science & Engineering* 47; Robert W Howarth, Anthony Ingraffea and Terry Engelder, “Natural Gas: Should Fracking Stop?” (2011) 477 *Nature* 271; See also International Energy Agency, n 26.

²⁹ Howarth, Ingraffea and Engelder, n 28.

³⁰ Howarth, n 28; Howarth, Ingraffea and Engelder, n 28.

³¹ Compare Climate Analytics, *Why Using 20-year Global Warming Potentials (GWPs) for Emission Targets Are a Very Bad Idea for Climate Policy* (2017) <<https://climateanalytics.org/briefings/why-using-20-year-global-warming-potentials-gwps-for-emission-targets-is-a-very-bad-idea-for-climate-policy/>>; International Energy Agency, n 26. In fact, the work of Howarth et al has been criticised on the grounds that it overestimates amounts of fugitive emissions and uses timescales that fail to capture the contrast between the long residence time of CO₂ in the atmosphere and the short residence time of methane: Lawrence M Cathles et al, “A Commentary on ‘The Greenhouse-Gas Footprint of Natural Gas in Shale Formations’ (2012) by R.W. Howarth, R. Santoro, and Anthony Ingraffea” (2012) 113(2) *Climatic Change* 525.

³² James Hansen et al, “Climate Change and Trace Gases” (2007) 365 *Philosophical Transactions. Series A, Mathematical, Physical, and Engineering Sciences* 1925; D Shindell et al, “Simultaneously Mitigating Near-term Climate Change and Improving Human Health and Food Security” (2012) 335 *Science* 183; United Nations Framework Convention on Climate Change, *Report on the Structured Expert Dialogue on the 2013-2015 Review* (2015) <<http://unfccc.int/resource/docs/2015/sb/eng/inf01.pdf>>. The United Nations' Structure Expert Dialogue notes that reaching the 2 degrees Celsius scenario is best construed as a critical threshold after which high-impact climate events are to be expected.

³³ David Archer and Bruce Buffett, “Time-dependent Response of the Global Ocean Clathrate Reservoir to Climatic and Anthropogenic Forcing” (2005) 6(3) *Geochemistry, Geophysics, Geosystems* 6.

³⁴ Ramon A Alvarez et al, “Greater Focus Needed on Methane Leakage from Natural Gas Infrastructure” (2012) 109 *Proceedings of the National Academy of Science* 6435.

GHG emissions against coal for electricity generation with reference to the rate of methane leakage. The rate of methane leakage is typically expressed as a fraction of the lifetime production of the gas well, normalised to the amount of methane in the gas produced.³⁵ The GHG emissions for using gas was found to be less than it was for coal only if the rate of methane leakage was less than 3.2%.

Of course, the conclusiveness of this evidence depends on how frequently we can expect a methane leakage rate of greater than 3.2%. In Australia, detailed estimates of actual measurements of fugitive emissions from unconventional gas are currently lacking (as are similar estimates for conventional gas³⁶). A further problem with methane emissions reporting in the Australian context is that it is not based on on-the-ground measurement, and is instead calculated using pre-determined factors. This method, which Australia employs in its national carbon budget reporting to the United Nations, has been questioned on the grounds of comprehensiveness and potential under-reporting by a number of sources.³⁷ Finally, Australia's reported methane emissions from the oil and gas sector (based on factors rather than actual measurements) are 10 to 20 times lower than recent methane leakage rates recorded in the atmosphere in the United States.³⁸

The most recent International Energy Agency (IEA) estimates provide a further indication of the uncertainty associated with gas measurements. The IEA³⁹ do paint a slightly more optimistic picture for gas: they estimate that gas has a global emissions intensity of 1.7% and claim that over a 100-year timeframe, if the global emissions intensity of gas is less than 5.5%, then gas has lower life cycle emissions than coal. They also claim that, over a 20-year timeframe, emissions intensity must be below 3% for gas to be cleaner than coal. However, the IEA notes that the data used has a very high level of uncertainty, both because the emissions intensity estimate is not based on data submitted by individual countries, but rather on extrapolations from the US Greenhouse Gas Inventory. The IEA⁴⁰ acknowledge that these will be inaccurate for at least some countries, given the great heterogeneity of methane leakage rates. They also note that the accuracy of the US Greenhouse Gas Inventory is questionable, with variations of compared emissions levels ranging from plus to minus 25%.⁴¹

Where do we stand then, with regard to the extent of the harm associated with gas emissions? First, there is at least some evidence suggesting that gas may be more harmful to the climate than has previously been supposed once full life cycle GHG emissions are taken into consideration.⁴² Second, the problems surrounding the uncertainty of measurements from methane leakage are universally acknowledged in these debates and make it difficult to estimate exactly how damaging gas is to the climate. It is beyond the scope of this article to attempt to arbitrate between competing perspectives on the exact level of harm posed by gas emissions and whether or not it is better than other fossil fuels under all scenarios.

³⁵ Howarth, n 28.

³⁶ Dimitri Lafleur et al, *Review of Methane Emissions* (2016) <http://energy.unimelb.edu.au/_data/assets/pdf_file/0019/2136223/MEI-Review-of-Methane-Emissions-26-October-2016.pdf>.

³⁷ Tim Forcey, *Infrared Video-recording Methane Emissions in the Queensland Coal Seam Gas Fields* (2017) <https://d3n8a8pro7vhmx.cloudfront.net/lockthegate/pages/3861/attachments/original/1495068484/FLIR_Report_FINAL_20170501_Small.pdf?1495068484>; New South Wales Chief Scientist, *Initial Report on the Independent Review of Coal Seam Gas Activities in NSW* (2013) <http://www.chiefscientist.nsw.gov.au/_data/assets/pdf_file/0016/31246/130730_1046_CSE-CSG-July-report.pdf>; United Nations Framework Convention on Climate Change, *Report on the Individual Review of the Inventory Submission of Australia Submitted in 2015, Note by the Expert Review Team* (2016) <<http://unfccc.int/resource/docs/2016/arr/aus.pdf>>. Forcey recently highlighted this point through an infrared video recording of an unquantified amount of methane released from a coal-seam-water pipeline vent in a Queensland coal seam gas field. The report noted that emissions of the type captured on video footage are effectively assumed to be zero from the perspective of global emissions reporting procedures.

³⁸ Lafleur et al, n 36, 64.

³⁹ International Energy Agency, n 26.

⁴⁰ International Energy Agency, n 26.

⁴¹ These variations concern year-on-year changes for 10 previously published Inventories compared against the latest estimates from the 2017 report: see International Energy Agency, n 26, 410.

⁴² Alvarez et al, n 34; Howarth, Ingraffea and Engelder, n 28; Lafleur et al, n 36.

Nonetheless, the studies above indicate that gas is likely to be a greater contributor to climate change than previously assumed.⁴³

B. The Harm of Opportunity Costs Associated with Gas Infrastructure

Before closing this section it is worth outlining an additional problem associated with gas that is separate from the “bare” or direct harm of its contribution to global warming. This is that there are a number of opportunity costs associated with investing in gas technologies, infrastructure, and markets.

The first concerns resources and financing. The more a country works to acquire infrastructure (eg pipelines to transport gas, power plants) and develop trade relationships for importing or exporting of gas, the less resources might be available for investment in renewable technologies. As an example, consider the production and transportation infrastructure required for gas and LNG. Transporting gas requires construction of physical pipelines, while transporting LNG requires building ships to keep it liquefied and facilities to transform it from a gas into a liquid. Producing gas for consumption requires production plants. In addition to these resources, those invested in the gas market will likely promote the development of resources that depend on gas, for example, new power plants that are designed to run on gas, automobile infrastructure developed to support gas vehicles, and so on. Development of gas infrastructure may then also lead to attempts to lengthen the life of the infrastructure for economic reasons, which will then create a competition among resources that may reduce the possibility of developing renewable technologies.⁴⁴ To the extent that these resources are invested in, the opportunity to invest in the development of renewable energy sources is lost. In this way, gas infrastructure can create “lock-in” effects. Lock-in effects occur when longer-lived investments in LNG distribution systems prevent the deployment of resources for investing in lower-carbon alternatives, and when such existing resources are less well-suited than power plants for carbon capture and storage.

The second issue concerns energy consumption and energy efficiency. Several global studies using energy-economy models to reflect interactions between energy supplies, prices and consumption show that more abundant, inexpensive gas supplies can lead to increased energy consumption, partly or fully offsetting the GHG benefits from substitution of other fuels.⁴⁵ This is known as the “scale” effect: to the extent that added gas supplies lead to lower energy prices, economic activity is increased, promoting more energy-intensive activities, and reducing incentives for energy efficiency. These effects in turn can result in increased energy use and CO₂-e emissions. The scale effect has been noted for several years,⁴⁶ but has only more recently come into focus with modelling exercises that investigating the market-driven impact of more abundant gas supplies. In this way, abundant gas supplies threaten to increase net emissions, in virtue of reducing incentives for energy efficiency. This will also affect demand for innovative approaches to improving energy efficiency.

In summary, even if gas has some utility as a transitional bridge fuel from an all-things-considered perspective, it remains a significant contributor to the threat of climate change. It should be noted that the argument made here can be made independently of considerations relating to the negative effects of unconventional gas extraction methods (especially hydraulic fracturing technologies) on human health and the environment (considered apart from global warming-related considerations), of which many have

⁴³ Comparing different categories of harms and benefits from an all-things-considered perspective is an important enterprise, but one orthogonal to the main purpose of this article. This article focuses on connecting the gas export industry to climate harms in particular. Maintaining energy security is an important consideration that may motivate a limited role for gas in the near future (compare International Energy Agency, n 26; Benjamin K Sovacool, RV Sidortsov and Benjamin R Jones, *Energy Security, Equality and Justice* (Routledge, 2013)), but the focus here is on the level of moral responsibility agents must take for their involvement in specifically climate-related harms.

⁴⁴ Ben Caldecott et al, *Stranded Assets: A Climate Risk Challenge* (2016) <<https://publications.iadb.org/bitstream/handle/11319/7946/Stranded-Assets-A-Climate-Risk-Challenge-Monograph.pdf?sequence=1>>.

⁴⁵ Michael Lazarus et al, *Natural Gas Guardrails Climate Bridge* (2015) <http://sei-us.org/Publications_PDF/NCE-SEI-2015-Natural-gas-guardrails-climate-bridge.pdf>; H McJeon et al, “Limited Impact on Decadal-scale Climate Change from Increased Use of Natural Gas” (2014) 514 *Nature* 482.

⁴⁶ S Brown, A Krupnick and M Walls, *Natural Gas: a Bridge to a Low-Carbon Future* (Resource for the Future, 2009) <<http://www.rff.org/RFF/Documents/RFF-IB-09-11.pdf>>.

been well-documented. There are concerns over the chemicals used in fracturing fluids, with potential for water, air, and soil pollution.⁴⁷ There is also evidence that these forms of pollution have deleterious effects on human health. Through skin contact, respiration, or ingestion, contaminants originating from unconventional gas extraction methods can lead to a wide array of health problems, ranging from premature births to cardiac arrest.⁴⁸ Though it is not the place here to exhaustively catalogue these effects, a recent meta-analysis found that more than 80% of studies looking at the health impacts of unconventional gas development and production documented positive associations between either risks or actual harms.⁴⁹ Though the fact that gas use produces harmful GHGs is sufficient to generate a moral obligation to not use it as a bridge fuel, when the harm of emissions is considered in conjunction with deleterious health and environmental effects, the case supporting moral liability is stronger still.

V. POLICY IMPLICATIONS

We have argued that Australia has a moral obligation to bear responsibility for its complicity in climate harms arising from its substantial and growing gas export industry. This industry renders Australia complicit in: (1) the harm of GHG emissions from burning of gas and gas extraction; and (2) opportunity costs associated with a delayed transition to renewable technologies.

But exactly what form should accepting moral responsibility for complicity in these harms take? Three options are mentioned here as a starting point for thinking about how to use the notion of complicity to motivate new policy mechanisms. These options are: (1) adjusting Australia's emissions reductions targets; (2) constraining exports; and (3) increasing funds to global mitigation efforts.

The first option is to "offset" harm by accepting responsibility for the consequences of complicity in harm, through for instance adjusting Australia's emissions reductions targets. For example, as part of its COP21 Paris commitments, Australia has agreed to reduce its annual emissions by 26–28% below 2005 levels by 2030, which builds on its 2020 target of reducing emissions by 5% below 2000 levels. The 2020 target is 532 Mt CO₂-e and the 2030 target is 440–452 Mt CO₂-e.⁵⁰ Understanding these targets is important if the significance of continuing to produce and export gas is to be properly appreciated. Even though these targets are likely to be too low to adequately address Australia's fair share,⁵¹ if we add the contribution gas exports make to these commitments, Australia's pledges are even more problematic. If projected gas export emissions for the year 2020–2021 are added to projected national emissions from direct consumption for 2020–2021, this yields a total of 861.34 MtCO₂-e, which is roughly one-third above the stated target.⁵² While one might not think that Australia ought to bear 100% of the responsibility for emissions associated with gas exports, one might nonetheless think that recognising complicity requires that Australia adopt responsibility for *some* proportion of the relevant emissions (such as 50% or 25%). The case for bearing responsibility in this way is strengthened by the fact that

⁴⁷ Jeremy Moss, Alicia Coram and Grant Blashki, *Is Fracking Good for Your Health? An Analysis of the Impacts of Unconventional Gas on Health and Climate* (The Australia Institute, 2013).

⁴⁸ Moss, Coram and Blashki, n 47.

⁴⁹ Jake Hays and Seth BC Shonkoff, "Toward an Understanding of the Environmental and Public Health Impacts of Unconventional Natural Gas Development: A Categorical Assessment of the Peer-reviewed Scientific Literature, 2009-2015" (2016) 11 *PLOS ONE* 1–13.

⁵⁰ Australian Government, Department of the Environment and Energy, n 17.

⁵¹ A recent statement by the Climate Council (Tim Flannery et al, *Halfway to Paris: How the World Is Tracking on Climate Change* (2015) The Climate Council <<https://www.climatecouncil.org.au/uploads/7ec258783dd2367efa806f6dc6c9a54a.pdf>>) claims that Australia must cut its GHG emissions much more deeply and rapidly to contribute its fair share in meeting the climate change challenge. A 2030 target of a 40–60% reduction below 2000 levels (or a range of approximately 45 to 65% below 2005 levels) is the bare minimum for Australia to be both in line with the science and the rest of the world.

⁵² It should be borne in mind that the figures stated here follow the reporting guidelines of the Intergovernmental Panel on Climate Change: Intergovernmental Panel on Climate Change, *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (2014); Intergovernmental Panel on Climate Change, *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change* (2014). Here, radiative forcing for a GHG is calculated across a 100-year period. As such, these figures provide estimates lower than what would be provided using a 20-year GWP approach.

Australia has been under scrutiny in the global community for not taking on its “fair share” of the burden of mitigating climate change at the global scale.⁵³ No matter what the final increased carbon budget is, it will require the transition to a low-carbon society to be even more comprehensive.

Another way Australia might accept moral responsibility for complicity is by simply stopping its contribution to the harm in which it is complicit via adopting “supply side” measures. As the name suggests, supply side measures focus on altering the supply of fossil fuels. Supply side policies can be characterised as either restrictive – constraining or preventing supply in some way (production taxes/quotas) – or supportive (provision of renewable energy/feed in tariffs).⁵⁴ The measures that are used to achieve either of these goals are varied. They range from economic measures such as production taxes, export taxes, removal of producer subsidies to regulatory restrictions, such as the imposition of quotas, the closure of old mines, or limiting the use of state-owned resources such as water or land. Supply side measures could also involve reducing the value of companies and access to capital via divestment from such companies. Local and national governments or non-state actors might adopt policies that banned their institutions from investing in fossil fuel supply chains⁵⁵ and simultaneously encouraging them to invest or provide finance to non fossil fuel resources.⁵⁶

The most direct way Australia could do this would be to simply stop allowing and supporting companies to continue extracting and exporting gas. For instance, Australia could commit to no longer granting licenses for new gas projects, it could enact new laws that forbid or restrict the extraction of gas, and it could stop providing subsidies to companies extracting and exporting gas. For instance, Australia could commit to phasing out exports of gas over a multi-year period, and could immediately stop granting licenses for new developments. This would also encourage companies currently invested in fossil fuel markets to move toward renewable technologies for energy supply. This would in turn avoid opportunity costs associated with investing in infrastructure for fossil fuels.⁵⁷ Australia could also apply increasingly stringent taxes for gas, including import taxes, and taxes for producers of exports and developers of new gas facilities, while still moving towards a complete phasing out of gas extracting and exporting.

The advantage of supply side measures, especially restrictive ones, is that they target the agents who are arguably highly responsible for the threat of climate change and any harms already caused and indicate how costs ought to be distributed. Supply side responses such as mine gate taxes, production quotas or restrictions on water/land use impact upstream producers of fossil fuels directly as opposed to measures that target the final users of their products. Fossil fuel producers, in particular the “carbon majors” (large fossil fuel producers to whom supply side measures are likely to apply) bear a high degree of responsibility for the harms caused by climate change.

This is not to deny that the threat of climate change has multiple types of agents who have made and continue to make a substantial and morally relevant contribution. National and local governments, consumers and other businesses also arguably share the blame. However, the role of fossil fuel producers, especially the carbon majors, in significantly contributing to climate change have been extensively discussed.⁵⁸

⁵³ Climate Action Tracker, *Australia Climate. Action Tracker* (2016) <<http://climateactiontracker.org/countries/australia.html>>; Climate Council, *Towards Morocco: Tracking Global Climate Progress Since Paris* (2016); Robyn Eckersley, “The Politics of Carbon Leakage and the Fairness of Border Measures” (2010) 24(4) *Ethics & International Affairs* 367; Bill Hare et al, *The Finkel Review and Scientific Consistency with Paris Agreement* (2017) <http://climateanalytics.org/files/finkel_review_and_scientific_consistency_with_paris_agreement.pdf>; Jeremy Moss, “Mining, Morality and the Obligations of Fossil Fuel Exporting Countries” (2016) 51 *Australian Journal of Political Science* 496.

⁵⁴ F Green and R Dennis, “Cutting with Both Arms of the Scissors: The Economic and Political Case for Restrictive Supply-side Policies” (2018) *Climatic Change*.

⁵⁵ J Moss, “The Morality of Divestment” (2017) 39(4) *Law and Policy*.

⁵⁶ M Lazarus and H Van Asselt, “Fossil Fuel Supply and Climate Policy: Exploring the Road Less Taken” (2018) *Climatic Change*.

⁵⁷ Lazarus et al, n 45.

⁵⁸ B Ekwurzel, et al. “The Rise in Global Atmospheric CO₂, Surface Temperature, and Sea Level from Emissions Traced to Major Carbon Producers” (2017) 144 *Climatic change* 579–590; Richard Heede, “Tracing Anthropogenic Carbon Dioxide and Methane Emissions to Fossil Fuel and Cement Producers, 1854–2010” (2014) 122 *Climatic Change* 1–2.

A final suggestion for action could take the form of Australia increasing its contribution to global mitigation funds in line with increased responsibility for exported emissions. On the basis of its Paris commitments Australia has an obligation to reduce its domestic emissions. Australia has also agreed to contribute to other countries' mitigation efforts via various mechanisms, including the Green Climate Fund, which was set up to address the pressing mitigation and adaptation needs of developing countries. Australia has pledged US\$200 million between 2015 and 2018.⁵⁹ This amount could be increased in proportion to the liability for exported gas emissions.

VI. CONCLUSION

If the global community increases or retains its reliance on gas, a reduction in the levels of GHG emissions necessary to avoid exceeding the global mean temperature increase limit required to avoid dangerous climate change will become increasingly unlikely. This article has connected large gas-exporting industries to this risk, arguing that Australia, as the world's second-largest exporter of gas, has a moral obligation to mitigate harms arising from this industry. Doing so will require either a rapid phasing out of the gas export industry altogether, or accepting responsibility for the consequences for the climate of its export industry. Steps could be taken simultaneously, both towards phasing out the industry, while accepting responsibility for consequences while this transition occurs. This argument also applies to other countries engaged in exporting significant amounts of harmful fossil fuels: to the extent that a country is complicit in such harms, they should be held responsible, by stopping their contribution to such harms, by accepting some responsibility for the consequences thereof, or both. An application of this principle across the board is consistent with an outward-looking approach to mitigating global warming – that is, an approach that focuses not exclusively on GHGs produced from the energy consumption of individual countries, but on the ways a country's energy portfolio influences the global energy marketplace, and how this marketplace in turn constrains how quickly the global community can transition to a fossil fuel-free future.⁶⁰

⁵⁹ Julie Bishop, Minister For Foreign Affairs, Australian Department of Foreign Affairs and Trade, *Australia to Lead Green Climate Fund Board in 2017* (Media Release, 16 December 2016) <https://foreignminister.gov.au/releases/Pages/2016/jb_mr_161216.aspx>.

⁶⁰ Australian Bureau of Statistics, *Population clock* (2017) <<https://www.abs.gov.au/ausstats/abs@.nsf/0/1647509ef7e25faaca2568a900154b63?OpenDocument>>; Christian Barry and Gerhard Øverland, "Individual Responsibility for Carbon Emissions: Is There Anything Wrong with Overdetermining Harm" in J Moss (ed), *Climate Change and Justice* (CUP, 2015); Andrew Blakers, Bin Lu and Matthew Stocks, "100% Renewable Energy in Australia" (2017) 133 *Energy* 471; AR Brandt et al, "Methane Leaks from North American Natural Gas Systems" (2015) 343 *Science* 733; Andrew Burnham et al, "Life-Cycle Greenhouse Gas Emissions of Shale Gas, Natural Gas, Coal, and Petroleum" (2012) 46 *Environmental Science and Technology* 619; Carbon Brief, *Analysis: Just Four Years Left of the 1.5C Carbon Budget* (2017) <<https://www.carbonbrief.org/analysis-four-years-left-one-point-five-carbon-budget>>; Climate Action Tracker, *Australia Set to Overshoot Its 2030 Target by Large Margin* (2015) <http://climateactiontracker.org/assets/publications/briefing_papers/082015_Australia.pdf>; CSIRO, *Energy Networks Australia, Electricity Network Transformation Roadmap: Key Concepts Report* <http://www.energynetworks.com.au/sites/default/files/key_concepts_report_2016.pdf>; Robert W Howarth, *Preliminary Assessment of the Greenhouse Gas Emissions from Natural Gas Obtained by Hydraulic Fracturing* (2010) <https://s3.amazonaws.com/files.technologyreview.com/p/pub/legacy/ghg_emissions_from_marcellus_shale_april12010_draft.pdf>; Nathan Hultman et al, "The Greenhouse Impact of Unconventional Gas for Electricity Generation" (2011) 6 *Environmental Research Letters* 044008; International Energy Agency, *Are We Entering a Golden Age of Gas?* (2011) <http://www.worldenergyoutlook.org/media/weowebsite/2011/WEO2011_GoldenAgeofGasReport.pdf>; Sanford H Kadish, "Complicity, Cause and Blame: A Study in the Interpretation of Doctrine" (1985) 73 *California Law Review* 323; International Energy Agency, *Medium-term Gas Market Report 2013* (2013) <http://www.iea.org/publications/freepublications/publication/MTGMR2013_free.pdf>; Mercator Research Institute on Global Commons and Climate Change, *Remaining CO2 Budget* (2017) <<https://www.mcc-berlin.net/en/research/co2-budget.html>>; Benjamin K Sovacool, "Cornucopia or Curse? Reviewing the Costs and Benefits of Shale Gas Hydraulic Fracturing (Fracking)" (2014) 37 *Renew: Sustainable Energy Review* 249; Mathew Stocks, Andrew Blakers and Bin Lu, *Pumped Hydro Energy Storage to Support a 50–100% Renewable Electricity Grid* (2017) <<http://energy.anu.edu.au/files/Senate%20submission%20-%20ANU%20pumped%20hydro%20energy%20storage%20030217.pdf>>; United States Census Bureau, *U.S. and World Population Clock* (2017) United States Census Bureau <<https://www.census.gov/popclock/>>; Tom ML Wigley, "Coal to Gas: The Influence of Methane Leakage" (2011) 108 *Climatic Change* 601; World Resources Institute, *Greenhouse Gas Protocol Product Life Cycle Accounting and Reporting Standard* (2014) <<http://www.wri.org/publication/greenhouse-gas-protocol-product-life-cycle-accounting-and-reporting-standard>>.